

G. C. E. (Adv. Level) Examination - 2027

Combined Mathematics

Model Paper - 04

❖ All questions should be answered

2 Hours

1)

- a. α and β are roots of the equation $x^2 - px + q = 0$.
Find the equation whose roots are $\alpha(\alpha + \beta)$ and $\beta(\alpha + \beta)$.
- b. Find the values of λ for which the expression $f(x, y) = 2x^2 + \lambda xy + 3y^2 - 5y - 2$ can be written as the product of two linear factors

(30 min)

2)

- a. Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos^2(2\sin x)}{1 - \cos 2x}$
- b. If a and b are positive real numbers, show that

$$\log_a b = \frac{1}{\log_b a}.$$

Show that

$$\frac{1}{\log_2 2027} + \frac{1}{\log_3 2027} + \frac{1}{\log_4 2027} + \dots + \frac{1}{\log_{100} 2027} + \frac{1}{\log_{100} 2027}$$

- c. Solve the equation

$$4 \sin^2 x + 12 \sin x \cos x - \cos^2 x + 5 = 0 \text{ for } 0 \leq x \leq 2\pi$$

(30 min)

- 3) A train, 100 metres long, starts from rest at a station A and moves with constant acceleration. Later, it takes 10 seconds to pass a signal post B. The train is moving with velocity 11 ms^{-1} when the rear of the train passes B. Draw a velocity -time graph for the motion of the train.

Using this graph or otherwise,

- i. find the velocity with which and the time at which the front of the train passed B.
- ii. find the acceleration of the train and show that the total distance travelled is 302.5 metres, when the rear of the train is at B.

(15 min)

- 4) A system of coplanar forces consists of three forces (measured in newtons) acting respectively at points specified as follows:

Point	Position Vector	Force
A	$2j + 5j$	$P(i+3j)$
B	$4j$	$-P(2i+j)$
C	$-i + j$	$P(i-2j)$

Here i, j denote unit vectors along rectangular Cartesian axes Ox, Oy respectively, the unit of length being the metre.

Mark these forces, in component form, in a representative diagram, indicating the coordinates of the respective points of application.

Hence or otherwise, show that the system is equivalent to a couple of moment $10P \text{ Nm}$, and indicate the sense of the couple.

D is the point with position vector $2i$. The given system of forces can be brought to equilibrium by introducing three additional forces along the sides of the triangle OAD , taken in order and proportional to their lengths. Find these forces in vector form.

(15 min)

5) State the Sine Rule and the Cosine Rule for a triangle.

It is given that $\frac{b+c}{2k-1} = \frac{c+a}{2k} = \frac{a+b}{2k+1}$

where k is a given integer greater than 2 but not equal to 4 and a, b, c are the sides of a triangle ABC in the usual notation.

Show that $\frac{\sin A}{k+1} + \frac{\sin B}{k} = \frac{\sin C}{k-1}$

Also obtain $\cos A$ in term of k and show that

$$\frac{\cos A}{(k-4)(k+1)} = \frac{\cos B}{k^2+2} = \frac{\cos C}{(k+4)(k-1)}$$

where A, B, C have the usual meanings.

(15min)

Every question you attempt to answer is progress
Recheck calculations and final answer carefully